

Human-Mediated Sensing for AI Decision-Making under Uncertainty

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Abstract

The performance of AI agents in organizations and infrastructure is increasingly constrained not by computation, but by access to information absent from databases or APIs. Humans already serve as implicit sensors (e.g., answering queries, performing tests, and reporting observations), however existing systems treat this input as incidental rather than as an explicit, optimizable information source. We propose a framework that formalizes humans and human-mediated tests as explicit *sensing actions*, or a “human API,” for agentic systems. The model characterizes how human input reduces uncertainty, informs downstream decisions, and interacts with institutional incentives, while preserving human control over information flow. Applications to clinical decision support, institutional coordination, public health, and climate risk reveal counterintuitive phenomena, including regimes where additional input degrades performance and scenarios where selectively withholding information improves outcomes. Collectively, this provides a unified framework to integrate human intelligence into AI decision processes.

1 Introduction

Modern AI agents act not only on text, but through tools, APIs, and workflows. A customer support agent queries ticketing systems and CRMs [10], a clinical AI assistant reads electronic health records [35], and a planning agent calls scheduling and routing services [30]. In each case, the agent’s view of the world is mediated through a set of machine-readable interfaces.

Yet much of the state that matters for good decisions is not present in any existing API. Physical conditions (e.g., “is there standing water at this bridge?”), tacit knowledge (e.g., “is this project politically sensitive?”) and local social norms (e.g., “would this outreach campaign be considered respectful here?”) rarely live in a database. In practice, people fill these gaps [2]. Systems already prompt users for additional information, and humans routinely act as the only route to uninstrumented parts of the world [6, 8].

A natural objection, then, is that nothing more needs to be said: agents already ask questions, and humans already respond. Our central claim is that this view misses something important. There is a crucial difference between an agent that *implicitly* asks questions as part of a conversation, and one that treats human input as a set of *explicit sensing actions* with costs, constraints, and value-of-information [15, 21, 29, 33]. Without the latter, we cannot reason about, optimize, or govern how agents use humans as sensors (see Figure 1).

We propose to treat humans (and human-mediated tests) as explicit sensors [19, 31] that agents can call when tools are insufficient. The canonical example is clinical decision-making. A diagnostic agent can already do a great deal with data from an electronic health record and its pretraining prior [14]. But certain decisions require additional information that can only be

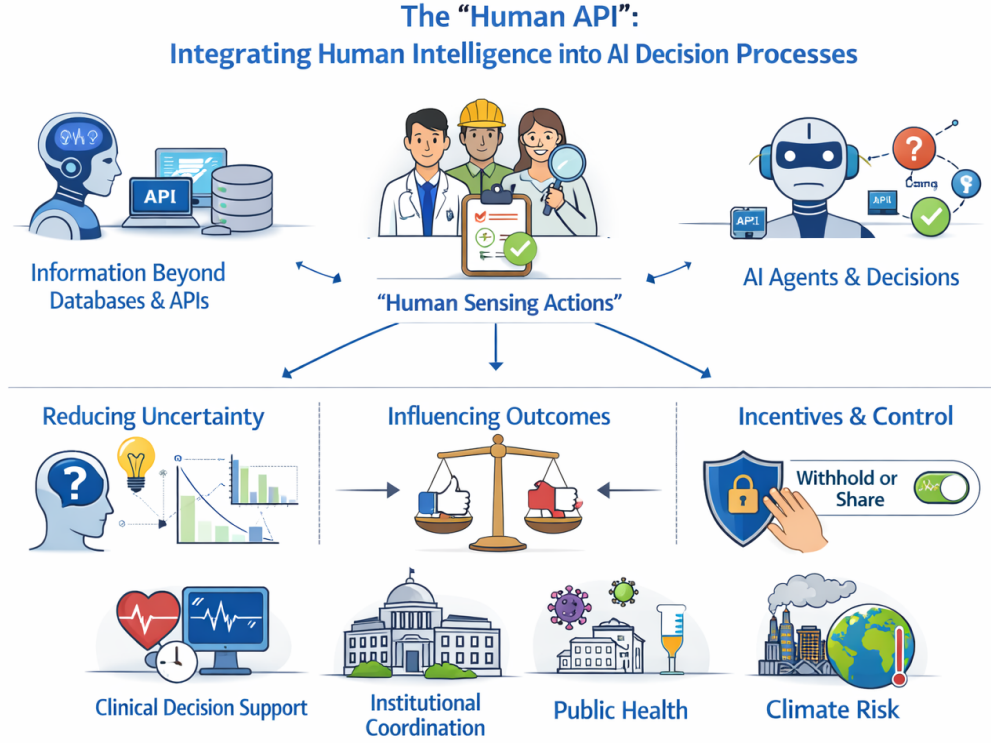


Figure 1: The figure illustrates the proposed *Human API* framework, wherein human actors perform explicit sensing actions, such as observations, tests, and measurements, that are integrated into AI decision-making processes.

obtained if someone asks the patient a question, performs an examination, or orders an expensive test such as a CT scan [16].

In our framework, these are not vague “clarifying questions” but distinct sensing actions with associated costs and expected benefits. Some are cheap and low risk (asking the patient where the pain is, checking for leg swelling), others are expensive, risky, or burdensome (imaging, invasive procedures). The agent’s problem is to decide *which* sensing actions to take, *when*, and *through whom*, given its current uncertainty and the costs of acquiring more information [12, 15, 37].

Concretely, in this paper, we:

1. Propose a compact model of agentic decision-making with explicit human sensing actions (a human API) and associated costs.
2. Use this model to characterize when human-mediated information helps or harms performance, leading to several counterintuitive phenomena.
3. Ground the framework in concrete scenarios from healthcare, institutional coordination, public health, and climate risk management.
4. Articulate design and governance implications for controlling what information humans provide to agents and under what incentives.

2 Motivating archetype: the clinician and the diagnostic AI assistant

We anchor our formalism in a setting that is already well understood: clinical decision-making in a teaching hospital [9, 29]. A clinician does not observe the true state of a patient directly. Instead, they form a hypothesis about what is going on by combining:

1. Existing records: vitals, medication lists, prior imaging reports, and notes.
2. Targeted questions: where the pain is, when it started, what makes it better or worse.
3. Tests and examinations of varying cost and invasiveness: physical exam, laboratory panels, and imaging.

They stop acquiring information once they believe they have enough to choose a good action (treatment, discharge, referral), given the costs and risks of further tests.

Now suppose we add a diagnostic AI assistant. The AI assistant ingests the electronic health record and basic vitals and produces a ranked list of candidate diagnoses with rationales and suggested next steps [28, 35]. For routine cases, this may be enough. For harder cases, the AI assistant will identify specific information that, if obtained, would sharply reduce uncertainty. For example:

My top hypotheses are pulmonary embolism and anxiety-related hyperventilation. I am uncertain because I do not know whether there is lower-limb swelling and how distressed the patient appears at the bedside. If you examine the patient and report (i) presence of swelling and (ii) a simple 1-5 rating of distress, I can refine my recommendation.

In this interaction, the junior doctor is acting as a sensor. They perform a low-cost sensing action (a brief exam) and return a small vector of features that the AI assistant could not observe otherwise. The AI assistant uses these features, together with its prior and the existing record, to update its beliefs and choose an action.

The same pattern arises for expensive tests. In some cases, the AI assistant can recommend a plan with reasonable confidence from existing information. In others, it will recognize that without the result of a CT scan, it cannot distinguish between conditions that require very different treatments. The CT scan is a high-cost sensing action: it is expensive, takes time, and exposes the patient to radiation, but it can dramatically reduce uncertainty. The AI assistant’s problem is not simply “can I think of a test?” as modern models can enumerate many tests, but “given my current uncertainty, is this test worth its cost?”

We now formalize this pattern in a way that extends naturally beyond medicine.

3 Model

In this section, we introduce a probabilistic framework to study humans as information-gathering sensors. Our goal is to rigorously capture how sequentially acquired observations influence decision-making under uncertainty. This approach allows us to quantify the value of human-mediated sensing, link it to Bayesian belief updates, and situate the problem within the framework of Markov decision processes (MDPs) [5, 18, 27]. By doing so, we can analyze both theoretical limits of information acquisition and practical considerations such as sensing costs and institutional incentives.

3.1 Probabilistic Setting

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space, and let $(\mathcal{S}, d)$ be a Polish space equipped with its Borel σ -algebra $\mathcal{B}(\mathcal{S})$. Let $\mathcal{P}_2(\mathcal{S})$ denote the space of probability measures on $\mathcal{S}$ with finite second moment. The underlying state of the world is a measurable random variable $S : \Omega \rightarrow \mathcal{S}$ with $\mu_0 := \mathbb{P} \circ S^{-1} \in \mathcal{P}_2(\mathcal{S})$. The measure μ_0 encodes the *prior uncertainty* over S , providing a formal starting point for inference. Working in a Polish space with finite second moments allows the use of Wasserstein geometry to quantify uncertainty reduction and stability under Bayesian updates [26, 36].

Uncertainty, Reward, and Sequencing Considerations. A natural way to frame the problem is in terms of *uncertainty*: for a given context, how do we maintain a quantitative estimate of when a particular sensing action is likely to be informative or appropriate? This requires being explicit about the *reward function*, since the notion of optimality depends on what we value—cost, performance, reliability, or some combination thereof.

Another crucial aspect is the *sequencing of actions*. The order in which sensing actions are executed can substantially affect outcomes: for instance, one might perform two inexpensive actions sequentially to gradually reduce uncertainty, or instead take a single costly action that provides a larger information gain but may incur higher delay or risk [7]. Capturing these trade-offs formally is essential for designing policies that effectively balance information acquisition, cost, and time, and underscores the importance of modeling the sequential decision problem explicitly.

3.2 Observations

Observations are generated in two stages:

1. **Baseline observations:** Let $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ be a measurable space. A baseline observation X is drawn via the Markov kernel $K_X : \mathcal{S} \rightarrow \mathcal{P}(\mathcal{X})$ where $X \mid S \sim K_X(S)$ representing information available without active sensing.
2. **Action-dependent observations:** Let $\mathcal{J}$ be a finite set of sensing actions. For each $j \in \mathcal{J}$, let $(\mathcal{Z}_j, \mathcal{B}(\mathcal{Z}_j))$ be a measurable space, and let $K_j : \mathcal{S} \rightarrow \mathcal{P}(\mathcal{Z}_j)$ where $Z_j \mid S \sim K_j(S)$, with cost $c_j \geq 0$. Selecting action j produces an observation Z_j , which allows the decision-maker to update the posterior belief over S . This is the mechanism by which active sensing reduces uncertainty and informs subsequent decisions.

We assume conditional independence of observations given S : $(X, \{Z_j\}_{j \in \mathcal{J}}) \mid S$ are independent. This ensures each observation provides new information about S , enabling tractable Bayesian updating.

3.3 Information Structure

Sequential information acquisition is formalized via a filtration of observed data. We formally list the following relevant definitions to help us understand the information structure.

Definition 3.1 (Information Filtration). Let $t \geq 0$, and let $j_1, \dots, j_t$ denote the first t realized sensing actions. Define

$$\mathcal{H}_t := \sigma(X, Z_{j_1}, \dots, Z_{j_t}), \quad \mathcal{H}_0 := \sigma(X).$$

The filtration $(\mathcal{H}_t)_{t \geq 0}$ captures all knowledge available at step t . By conditional independence, each new observation adds information about S that is not redundant with prior observations. Crucially, this filtration allows the posterior to summarize all past information relevant for future decisions which we denote by $\mu_t := \mathbb{P}(S \in \cdot \mid \mathcal{H}_t)$.

3.4 Query and Action Policies

Policies govern both sequential sensing and terminal actions:

Definition 3.2 (Query Policy). A query policy is a sequence of measurable mappings

$$\pi_t : (\Omega, \mathcal{H}_t) \rightarrow \{\text{stop}\} \cup (\mathcal{J} \setminus \{j_1, \dots, j_t\}),$$

selecting each action at most once. The associated stopping time is

$$\tau := \inf\{t \geq 0 : \pi_t = \text{stop}\}, \quad \mathbb{P}(\tau < \infty) = 1.$$

Each sensing action j_t generates Z_{j_t} , which updates μ_t and informs π_{t+1} .

Definition 3.3 (Action Policy). An action policy is a measurable mapping

$$\pi_{\text{act}} : (\Omega, \mathcal{H}_\tau) \rightarrow (\mathcal{A}, \mathcal{B}(\mathcal{A})), \quad A := \pi_{\text{act}}(X, Z_{S_\tau}),$$

where $S_\tau := \{j_1, \dots, j_\tau\}$ is the set of realized actions. The terminal action depends on the posterior belief μ_τ , which summarizes all information acquired via sensing.

3.5 Objective and Baseline Value

Let $U : \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ be a measurable utility function with $\mathbb{E}[|U(A, S)|] < \infty$. For a joint policy $\Pi \ni \pi = (\{\pi_t\}, \pi_{\text{act}})$, the expected payoff is

$$J(\pi) := \mathbb{E}\left[U(A, S) - \sum_{j \in S_\tau} c_j\right].$$

Let Π denote the class of admissible policies, and define a baseline policy π_{base} acting only on X :

$$V_0 := \mathbb{E}[U(\pi_{\text{base}}(X), S)], \quad \Delta V := \sup_{\pi \in \Pi} J(\pi) - V_0.$$

Here, ΔV quantifies the expected improvement from human-mediated sequential sensing.

Remark 3.4. We note that more information is not always beneficial: sensing costs or poorly integrated human input can lead to $\Delta V < 0$, so that sparse querying outperforms aggressive sensing (Lemma H.2).

3.6 Posterior Belief and Bayesian Updating

Given observations (x, z_{S_τ}) , the posterior is

$$\mu_{x, z_{S_\tau}}(B) := \mathbb{P}(S \in B \mid X = x, Z_{S_\tau} = z_{S_\tau})$$

for $B \in \mathcal{B}(\mathcal{S})$.

Here, each sensing action $j \in \mathcal{J}$ produces an observation Z_j that updates the posterior: $\mu_{t+1} = \text{BayesUpdate}(\mu_t, j_{t+1}, Z_{j_{t+1}})$. The updated posterior μ_{t+1} then serves as a sufficient statistic for all future actions and terminal decisions.

3.7 Quantifying Uncertainty Reduction

We can measure uncertainty reduction using the 2-Wasserstein distance W_2 [23, 36]:

$$\mathcal{U}(\mu) := W_2^2(\mu, \nu), \quad \Delta \mathcal{U}(j) := \mathbb{E}[\mathcal{U}(\mu_X) - \mathcal{U}(\mu_{X, Z_j})],$$

where $\nu \in \mathcal{P}_2(\mathcal{S})$ is a reference measure. This captures the expected contraction of the posterior distribution from sensing action j [36]. We emphasize the use of the 2-Wasserstein distance for

its well-established geometry-preserving properties; see, for instance, its application to graph analysis in infinite-dimensional spaces [25], as well as related variants designed for comparing higher-order relational structures [24].

We highlight that an explicit human API enables agents to quantify and communicate the marginal value of human input. Given the framework established so far, we note that an agent can estimate, for a particular sensing action j : (a) How much it expects its uncertainty to decrease if z_j is observed. (b) How likely that uncertainty reduction is to change the recommended action. (c) How expected utility, measured with respect to U , will change relative to baseline, net of c_j . This makes it possible for an agent to say not just “I would like more information” but “if you perform this specific test, I expect to reduce the risk of a serious error by this amount, at this cost”. This is qualitatively different from ad hoc clarifying questions and supports more deliberate, negotiated use of human sensing.

3.8 Institutional Incentives and Human Oversight

Institutional rewards can be incorporated via

$$R_{\text{inst}} : \mathcal{A} \times 2^{\mathcal{J}} \rightarrow \mathbb{R}, \quad (1)$$

$$J_{\text{tot}}(\pi) := \mathbb{E} \left[U(A, S) - \sum_{j \in S_\tau} c_j + R_{\text{inst}}(A, S_\tau) \right]. \quad (2)$$

Human oversight can be modeled by restricting the policy class to $\Pi_{\text{adm}} \subset \Pi$, separating epistemic control from environmental dynamics. We now state the main result, which shows that optimal sensing and decision-making can be carried out solely on the basis of posterior beliefs.

Theorem 3.5 (Optimality and Consistency of Belief-Based Policies). *Let $(\mathcal{S}, \mathcal{F})$ be a measurable space, and let $S \sim \mu_0$ on $\mathcal{S}$. Let X and $\{Z_j\}_{j \in \mathcal{J}}$ satisfy $X \mid S \sim K_X(S)$, $Z_j \mid S \sim K_j(S)$, conditionally independent given S . At each discrete time t , the agent selects $j_t \in \mathcal{J}$ or stops and chooses $A \in \mathcal{A}$ with reward $U(A, S)$. Denote the posterior by $\mu_t := \mathbb{P}(S \in \cdot \mid Z_{j_1}, \dots, Z_{j_t})$. Then, the following statements hold:*

1. **Posterior sufficiency.** *For any admissible policy π , there exists a belief-based policy $\tilde{\pi}$ with $\tilde{\pi}_t = f_t(\mu_t)$, $\tilde{\pi}_{\text{act}} = f_{\text{act}}(\mu_\tau)$, for measurable functions f_t, f_{act} , such that $J(\tilde{\pi}) = J(\pi)$.*
2. **Existence of optimal policy.** *There exists a belief-based policy π^* attaining $J(\pi^*) = \sup_{\pi \in \Pi} J(\pi)$.*
3. **Belief-state MDP.** *Under belief-based policies, $(\mu_t)_{t \geq 0}$ is a Markov process on $\mathcal{P}(\mathcal{S})$, reducing the sequential sensing problem to an MDP with state space $\mathcal{P}(\mathcal{S})$, action space $\mathcal{J} \cup \{\text{stop}\}$, and transition kernel induced by Bayesian updating.*
4. **Consistency with increasing sensing actions.** *Let $J^*(n)$ denote the optimal expected reward when at most n sensing actions are allowed. Then $(J^*(n))_{n \geq 0}$ is non-decreasing, converges to the full-information value $J_\infty^* := \sup_{\pi \in \Pi} J(\pi)$, and satisfies the sublinear bound: $J_\infty^* - J^*(n) \leq \frac{C}{n+1}$ for all $n \geq 0$.*

Remark 3.6. We note that the proof sketch of Theorem 3.5 is deferred to H.1 which can be loosely interpreted as follows:

1. **Posterior sufficiency:** The posterior μ_t summarizes all past observations. Once μ_t is known, no additional information from the observation history $\mathcal{H}_t$ improves decision-making. Any policy based on the full history can thus be equivalently expressed as a function of μ_t alone.

2. **Existence of an optimal policy:** There exists a policy π^* that attains the maximal expected payoff $J(\pi)$. Moreover, π^* can be taken to be belief-based, depending only on μ_t , which enables solution methods such as dynamic programming in the posterior space.
3. **Belief-state reduction:** Under belief-based policies, the posterior sequence $(\mu_t)_{t \geq 0}$ is Markov on $\mathcal{P}(\mathcal{S})$. Consequently, the sequential sensing problem reduces to a Markov decision process with state space $\mathcal{P}(\mathcal{S})$, action space $\mathcal{J} \cup \{\text{stop}\}$, and transitions given by Bayesian updating.
4. **Consistency with increasing sensing actions:** Let $J^*(n)$ denote the optimal expected payoff with at most n sensing actions. Then $J^*(n)$ is non-decreasing in n and converges to the full-information value J_∞^* as $n \rightarrow \infty$. The convergence is sublinear: the marginal benefit of each additional sensing action decreases roughly as $1/n$, reflecting diminishing returns in sequential information acquisition.

3.9 Human control over the information channel

Finally, humans are not only sensors; they also control which sensing actions are actually executed. We can represent this by a human policy that maps proposed sensing actions to realized ones. Formally, let π_{human} denote a (possibly stochastic) policy under which a proposed sensor j may be: (a) Accepted as is, (b) Modified (for example, downgraded to a cheaper alternative), and (c) Rejected outright.

The realized set of sensing actions S and their observations z_S are thus co-produced by the agent’s query policy π_{query} and the human/institutional policy π_{human} . This interaction is central to our interest in *control over what information we provide to agents*. Even if an agent can articulate that a particular test would significantly increase its confidence, clinicians, patients, and institutions may judge that test inappropriate given cost, risk, or fairness considerations and decline to execute that sensing action. In later sections we discuss how explicit human APIs make such control easier to design, implement, and reason about.

4 Why explicit human APIs if agents already ask questions?

Large language models already ask follow-up questions in many applications. Chatbots request missing details, coding AI assistants ask for clarification, and medical bots prompt users to describe their symptoms. It is natural to ask why we need a separate notion of a “human API” when this behavior is already emerging. Our model offers three answers.

4.1 Questions as actions, not as conversation

First, sensing actions are not free. They incur costs in money, time, risk, and human attention. Implicit questions generated during a conversation are rarely treated as decisions with such costs. An explicit human API elevates certain questions and tests to *actions* in the agent’s policy class. Each sensing action: (a) has a name and semantics (e.g., “order CT chest with contrast” or “ask junior doctor to check for oedema”), (b) has an associated cost c_j , and (c) is subject to explicit institutional constraints and logging. Once sensing actions are treated as actions, we can design and evaluate query policies that balance information gain against cost, instead of relying on whatever questions emerge from a language model’s next-token predictions.

4.2 Fixed versus open action sets

Second, there is a difference between the agent’s ability to *propose* sensing strategies and its authority to *execute* them. In our framework, the system exposes a fixed (or at least structured) action set $\mathcal{J}$ of allowed sensing actions. The agent’s query policy selects from this set, and

proposals that fall outside $\mathcal{J}$ require explicit mapping or rejection. This separation allows LLMs to be creative in diagnosing informational gaps and suggesting what kind of information would be useful, while keeping execution constrained and auditable. Without it, the space of possible questions and tests is effectively unbounded: an unconstrained agent can continuously invent new, potentially inappropriate sensing behaviors.

4.3 Seeing and shaping incentives

Finally, explicit sensing actions and costs make visible how institutional incentives shape sensing behavior. When an agent optimizes $J(a, s, S)$ rather than $U(a, s)$ alone, its query policy may over-use sensors that increase R and under-use sensors that reduce it, regardless of their informational value. By making sensing explicit, we can: (a) detect systematic over-ordering of certain tests or over-burdening of particular groups as sensors, (b) introduce budgets and caps on sensing actions, and (c) redesign R and constraints so that optimal policies under J are better aligned with U . If human questions remain implicit, these patterns are much harder to see and correct.

The next section develop concrete scenarios and design principles using this framework, and show how it gives rise to several non-obvious phenomena around when humans-as-sensors help or hinder agentic systems.

5 Counterintuitive phenomena from the model

The core model also gives rise to several non-obvious phenomena. We highlight four that are especially relevant for the design of human APIs.

5.1 The upskilling-value paradox

As agents and humans interact over time, two coupled learning processes occur:

- The agent updates its parameters θ so that $\mathcal{P}_\theta(s | x)$ captures patterns it previously needed human features z_S to resolve.
- Humans update their own mental models by observing how the agent reasons and which features it requests.

An immediate consequence is that the marginal value of human sensing can *decrease* as collaboration proceeds (see Lemma H.3), even while human expertise and overall system performance improve. In terms of the model, V_0 increases as $\mathcal{P}_\theta(s | x)$ becomes more informative, so the difference ΔV from a fixed set of sensing actions may shrink. This suggests a natural role shift: over time, humans may move from being routine sensors towards focusing their sensing effort on genuinely hard cases and on oversight of the sensing policy itself.

5.2 The “hiding information helps” effect

Some human-accessible features are systematically harmful when provided to agents. They may be spuriously predictive in historical data, encode social biases, or trigger brittle heuristics. In such cases, revealing these features through the human API can worsen both utility U and fairness metrics, even though they increase the agent’s information in a narrow sense. Our framework allows humans and institutions to implement policies π_{human} that intentionally withhold certain classes of information from agents. When designed carefully, such policies can increase V_{human} relative to a naive policy that transmits all available information, and can align decisions better with normative goals. This underscores that control over *which* facts flow to agents is as important as control over *how much* information is provided which is formally proven in Lemma H.4.

Sequencing and Predictive Posteriors. The order in which sensing actions are executed can significantly affect outcomes. For example, one might take two inexpensive actions sequentially to gradually reduce uncertainty [11, 15, 21], or instead take a single costly action that takes longer but yields a larger information gain. Formally capturing these trade-offs requires explicitly modeling the *sequential nature of decision-making*, as the optimal policy depends not only on which actions are taken, but also on their order.

Crucially, estimating the expected value or informativeness of any action requires access to the *predictive posterior*—the distribution of potential observations conditional on current knowledge and the chosen action. In practice, this may involve either computing the posterior analytically, if feasible, or employing a tractable approximation. The predictive posterior serves as the basis for evaluating expected information gain, anticipated reduction in uncertainty, and the trade-off between action cost and expected reward.

5.3 Selection bias in sensing

Finally, human sensing is subject to selection effects [2]. Humans are more likely to respond to sensing requests when cases are interesting, high-stakes, or personally salient. As a result, the distribution over (x, z_S) that the agent observes during learning is often not representative of the full deployment distribution. If the agent adapts π_{query} and its internal models based on this skewed experience, it may become:

- Overconfident on routine cases that rarely attracted human sensing or correction.
- Overly sensitive to patterns that occur disproportionately in the subset of cases where humans chose to engage.

Recognizing this selection bias is a first step towards correcting it, for example by logging which sensing requests were declined and explicitly modeling human availability and attention in the learning process which has been formally validated in Lemma H.5.

6 Scenarios and design patterns

We now illustrate the framework with a set of scenarios. Each combines a narrative description in a concrete domain; a mapping back to the model: state s , baseline observations x , sensing actions $\mathcal{J}$, and costs c_j ; and design and governance questions that arise when we treat human input as an explicit sensing API.

6.1 Teaching hospital: junior doctors as high-value sensors

Consider a teaching hospital where each ward has access to a diagnostic AI assistant. The AI assistant ingests the electronic health record and current vitals (the baseline x) and proposes a ranked list of candidate diagnoses with suggested management. For many routine cases, the baseline information suffices to choose a reasonable action. For more ambiguous cases, the AI assistant identifies specific, missing features that would substantially reduce uncertainty. It might say, for instance:

“If you check for lower-limb swelling and ask the patient to describe their shortness of breath in their own words, I can better distinguish between a pulmonary embolism and anxiety-related hyperventilation.”

Here, the sensing actions are:

- A brief physical exam (presence or absence of oedema).
- A structured elicitation of symptom quality.

We can model these as $j_1, j_2 \in \mathcal{J}$ with small costs c_{j_1}, c_{j_2} (time and effort for the junior doctor) and informative observation distributions $P_{j_1}(z_{j_1} | s), P_{j_2}(z_{j_2} | s)$. Over time, interactions like this have two coupled effects:

- The AI assistant learns, across many cases, which human-provided features most often change its posterior and downstream recommendations. This refines both π_{query} and its internal $P_{\theta}(s | x)$.
- Junior doctors learn, by seeing when and why certain features matter, which questions and exams have high value-of-information. Their own clinical judgment improves [14, 35].

Design questions include:

- How sparse should π_{query} be? How many sensing actions per case are acceptable before sensing burden becomes too high?
- How should responsibility for accepting or rejecting suggested sensing actions be communicated and shared between junior and senior clinicians?

6.2 Institutional knowledge-sharing: routing cases to the right team

In a large hospital network, a coordination agent must decide where to route complex cases: which clinic, which consultant, which multidisciplinary team [3]. The baseline information x consists of rosters, credentials, service descriptions, and historical workload. This captures formal constraints, but misses much of the institutional state:

- Which consultants are particularly strong on certain rare conditions.
- Which clinics have invisible slack this week.
- Which teams are already overwhelmed by other initiatives.

The agent therefore uses sensing actions that query humans about this informal state. Examples include:

- Asking a senior coordinator to rate, on a small scale, how appropriate each of three clinics would be for a given case.
- Asking a consultant whether they are willing to take on an additional complex case in the coming week.

These sensing actions have moderate costs c_j (time and attention of senior staff), but can significantly improve routing decisions, especially when baseline information is ambiguous. Design questions here include:

- How to allocate sensing burden fairly across staff, rather than repeatedly querying the most knowledgeable individuals.
- How to use observed answers to refine a map of institutional expertise and capacity, so that future decisions rely less on ad hoc sensing.

6.3 Social norms and local context in public health campaigns

A public health agent is tasked with designing vaccination outreach campaigns across regions [32]. The baseline x includes demographic data, previous uptake rates, and coarse measures of information access. However, crucial aspects of social acceptability and local norms are absent:

- Whether door-to-door visits are considered intrusive.

- Whether particular symbols or colors have political connotations.
- How messages should be framed to be respectful.

Here, sensing actions query local health workers or community leaders. A sensing action might be “ask a trusted local worker to rate the acceptability of door-to-door outreach on a 1-5 scale in this neighborhood”. The cost c_j is the worker’s time and any social risk they incur; the observation z_j informs the design of campaigns in that region. Design questions include:

- How to ensure that local sensing is high-bandwidth enough to capture nuance, rather than reducing rich local knowledge to a handful of numerical ratings.
- How to avoid systematic under-sensing of communities that are already under-served or harder to reach.

6.4 Physical micro-sensing for climate and infrastructure risk

Finally, consider agents that manage climate and infrastructure risks in flood- and fire-prone regions. Baseline information x comes from satellites, weather stations, and historical models, providing coarse-grained risk estimates [22]. Fine-grained decisions, however, depend on local conditions:

- The exact water level at a small bridge that may soon be overtopped.
- The presence of dry underbrush near a particular power line.
- Whether informal dwellings along a riverbank are currently occupied.

Sensing actions here include targeted micro-tasks sent to residents and volunteers: “if you are near bridge B , please take a photo facing upstream and answer two questions about the water height”. The costs c_j aggregate attention, battery, data usage, and any physical risk incurred. Design questions include:

- How to structure sensing campaigns so that high-value observations are collected without continuously burdening the same individuals.
- How to distinguish between crisis-time sensing (where higher burdens may be justified) and routine sensing (where stricter constraints on c_j are appropriate).

Across these scenarios, the model provides a common language for describing human sensing actions, their costs, and their expected value, and for designing agents that use humans as sensors in a controlled and accountable way.

7 Experiments

In this section, we validate the framework with a simulation on clinical decision support (Figure 2). An AI agent is prompted to perform a medical diagnosis with minimal case information. It has access to human sensors that it must query to obtain the information needed to issue a confident diagnosis. We model it as a Human API with predefined sensing actions that the agent can call. Humans are simulated using LLMs that have access to ground-truth context, as if a human were in the real world, performed the sensing action, and returned the information. The sensing actions are available in a predefined registry, each with a fixed cost. The central empirical question is *does the human API improve diagnostic accuracy and cost, and if it does, how does the sensing action space affect it?* Additionally, we compare this approach to an implicit version, in which the agent can interact with a human in a conversational format to obtain the information.

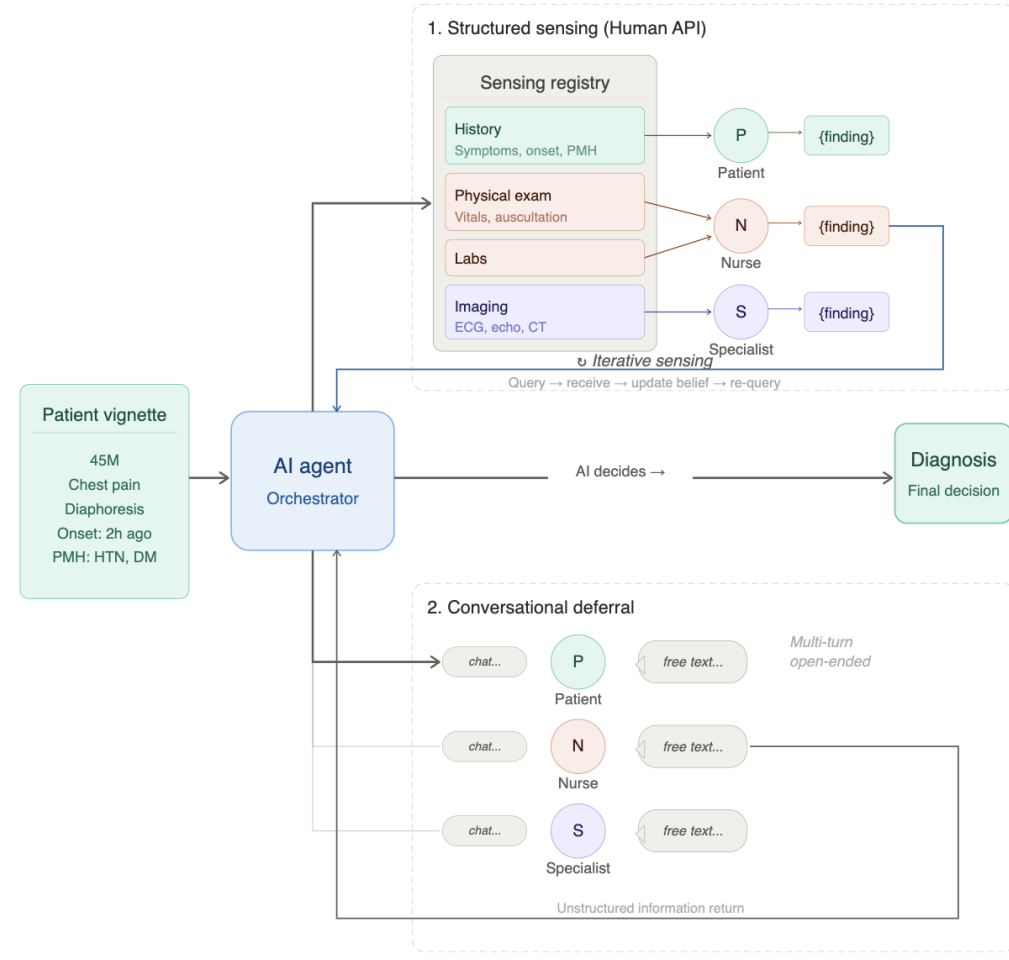


Figure 2: The humans-as-sensors framework. The AI agent receives a patient vignette and makes a diagnosis by querying humans either through structured sensing actions or conversational deferral.

7.1 Dataset

We evaluate the simulation using samples from the MedQA-USMLE-4-options dataset [17].

The dataset is a free-form multiple-choice dataset for medical problems. In our study, we sample from the 12,723 English examples. The subset contains vignettes with patient cases, and the questions ask for a diagnosis or treatment.

We use GPT-4.1 to preprocess the vignettes and extract patient information, such as age, gender, and chief complaint. This summary is provided to the agent as its baseline observation. The remainder is expanded into a rich patient profile covering demographics, complaints, vitals, physical exams, laboratory results, and imaging findings. The full profile serves as the ground truth that simulated human sensors can draw from when prompted to perform sensing.

To provide the agent with sensing tools, we construct a registry of 62 actions. Each action has a category (history, physical exam, labs, or imaging), cost, source (patient, nurse, or specialist), and response. The costs are fixed values. They are not intended to represent real-world prices. For instance, `hx_past_medical` (Past Medical History), of category History, costs 1 and is sourced from the patient; it returns a list of past conditions and procedures. The full registry of actions is available in Appendix A.

7.2 Main Comparison

We evaluate mean accuracy, cost, and turns on 582 cases and compare human sensing (H-API) and the conversational (Implicit) against the Baseline (model alone floor) and the Baseline-Full (model with full ground-truth context ceiling). Additionally, we compare the H-API across three levels of registry granularity, with less granular actions returning more information at a higher cost. Fine is a registry of 62 atomic sensing actions. Medium has 20 actions composed of fine. The coarse includes 8 broad categories. The mappings are documented in Table 6. The Implicit human is simulated with an LLM (GPT-4.1) that has access to the vignette. We also consider the case where the agent may need to converse with multiple humans (Implicit multi). The system prompts are available in Appendix B.

7.3 Information Analysis

To understand the two strategies in terms of information gathering, we perform two analyses. First, we classify the set of actions corresponding to each implicit turn with an LLM judge (GPT-4.1-mini) to compare Implicit and H-API on $N = 479$ shared cases to calculate coverage, Jaccard similarity, and novelty rate. Then, we study the per-turn information gain and uncertainty reduction by inserting diagnostic checkpoints at each sensing turn and forcing the models to make a prediction and using a separate scoring model to estimate the $P(\text{correct option} \mid \text{evidence so far})$ on a subset of 100 cases.

7.4 Complementarity Analysis

Human-AI collaboration is complementary when the team outperforms either party alone [4]; formally, $R(\pi^*) < \min(R_{\text{model}}, R_{\text{human}})$ under optimal deferral. We compare the agreement between H-API (fine) and Implicit (single) on 582 shared cases. For the 76 cases where they disagree, we attempt to pick the better one based on self-reported confidence and an independent LLM judge (GPT-4.1) that picks the most plausible diagnosis based on reasoning, and compare that to an oracle that always selects the better strategy.

7.5 Action-Space Scaling

We vary the sensing registry size from $k = 0$ to $k = 60$, randomly picking k actions with replacement, in 5 repetitions of 50 cases each per registry size, to study how the action space affects accuracy and cost. Drawing $k = 60$ from 62 actions yields ~ 39 unique actions in expectation (Appendix F).

Table 1: Collaboration strategies ($N = 582$ shared cases). Cost is mean sensing cost in registry points.

Condition	Acc. (%)	Cost	Turns
Baseline	71.8 ± 3.6	0.0	1.0
Baseline-Full	92.8 ± 2.0	54.2	1.0
H-API (fine)	85.1 ± 2.9	2.6	2.5
H-API (medium)	84.0 ± 2.9	7.0	2.2
H-API (coarse)	86.3 ± 2.6	13.6	2.1
Implicit (single)	86.8 ± 2.8	51.2	4.3
Implicit (multi)	83.5 ± 3.0	15.6	2.6

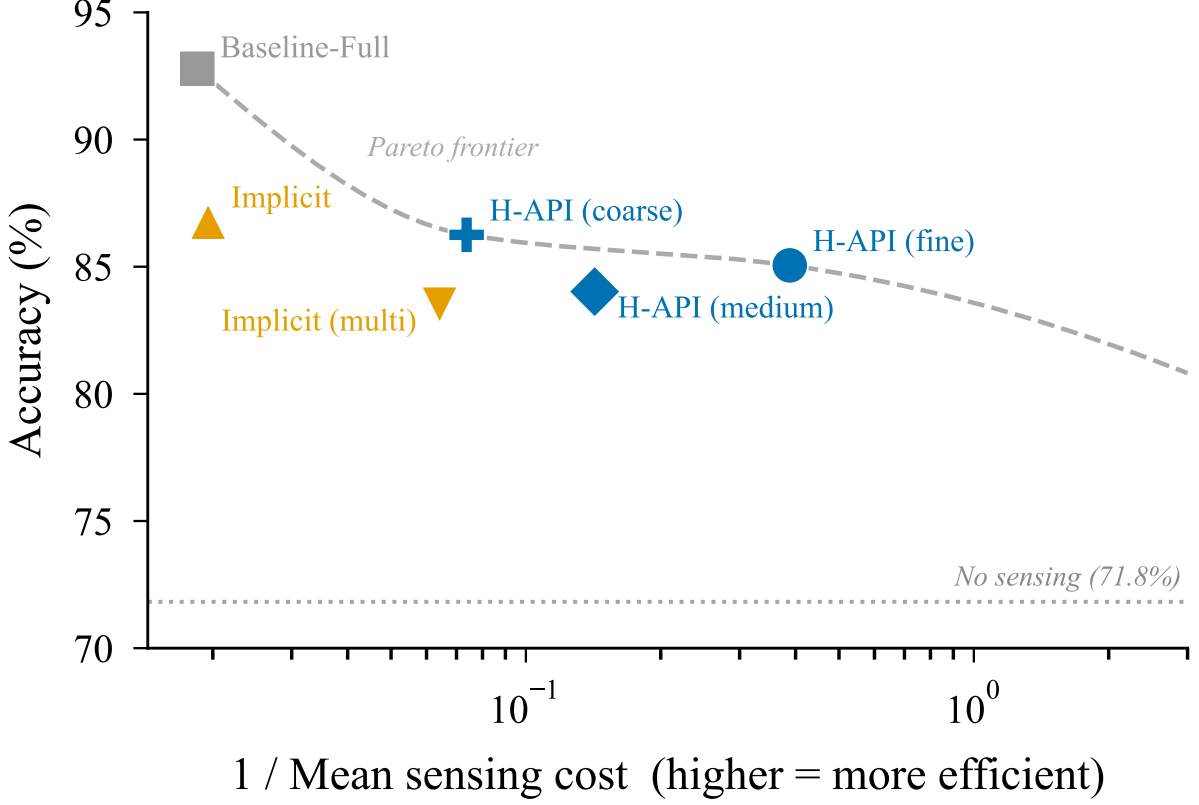


Figure 3: Cost-efficiency frontier. x -axis is $1/\text{cost}$ (higher = cheaper); dashed line is the Pareto frontier.

Table 2: Information overlap: implicit vs. H-API (fine) ($N = 479$).

Metric	Value
Impl. turns / case	3.7
Impl. actions / case	4.9
H-API actions / case	1.9
Coverage	0.478
Jaccard similarity	0.199
Novelty rate	2.7%

7.6 Results

7.6.1 Main Comparison

Table 1 compares all conditions showing that H-API and Implicit perform with comparable mean accuracy (83.5% – 86.8%), which improves over the Baseline (71.8%), but does not reach Baseline-Full (92.8%). H-API spent fewer resources than Implicit at all granularities, showing the benefit of structured sensing in requesting targeted cost-aware actions. Additionally, the trend shows the total cost increases as the registry size becomes coarser. The Pareto frontier (Figure 4a) passes through H-API (coarse) and H-API (fine) between the baselines. All conditions show overconfidence (ECE 0.200 baseline, 0.123 H-API fine), with the risk-coverage curve nearly flat (Figure 7, Appendix), motivating active sensing over confidence-based routing.

7.6.2 Information Analysis

The information analysis reveals that Implicit and H-API gather largely different information (Table 2; Jaccard similarity of 0.199). Implicit utilizes 4.9 actions on average compared to 1.9 for

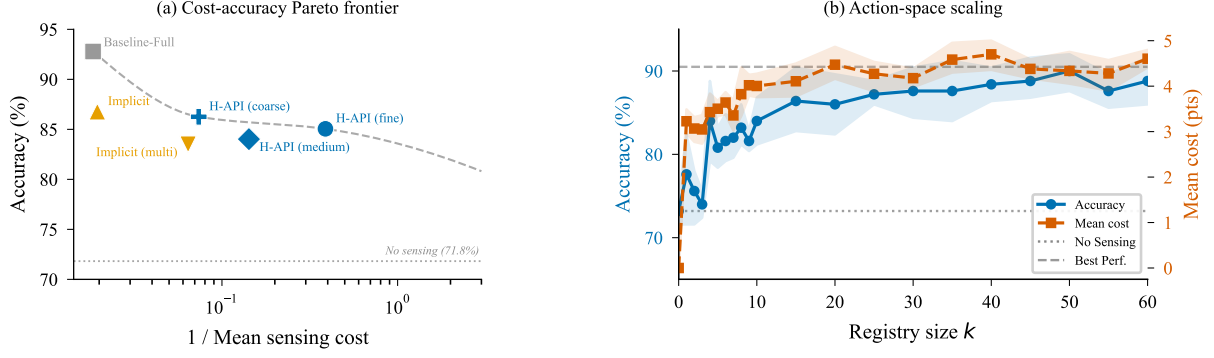
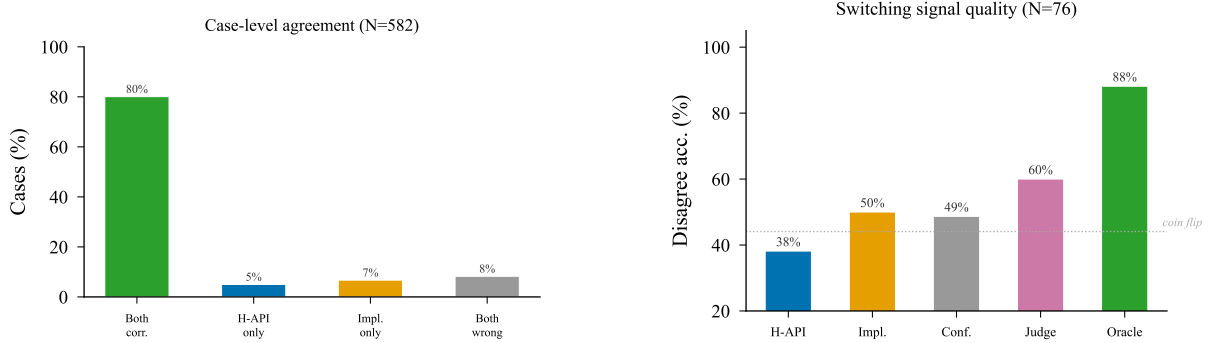


Figure 4: (a) Cost-accuracy Pareto frontier. (b) Action-space scaling. Accuracy (blue) and cost (red) vs. registry size k .

H-API. They also differ in how quickly they reduce uncertainty. Both start at $P(\text{correct}) \approx 0.34$ (Figure 6), but H-API increases steadily to 0.57 by turn 3. H-API entropy drops from 1.77 to 1.27 bits while Implicit’s plateaus near 1.4 bits after the first turn, suggesting diminishing returns from extended conversation. The small sample at later H-API checkpoints warrants caution.

7.6.3 Complementarity Analysis

The complementarity analysis reveals H-API and Implicit agree on 80.1% of cases, each uniquely solving 5–6% (Figure 5a), resulting in an oracle selector scoring 87.8% [85.2, 90.4]. In the cases where H-API and Implicit disagree, confidence-based switching scores 47% (slightly better than a coin flip at 44%) while the LLM judge reached 60%, improving over each independently but still far from the oracle selector (88%).



(a) Case-level agreement between H-API (fine) and Implicit (single). Each strategy uniquely solves 5–6% of cases.

(b) Switching signal quality on disagreement cases.

Figure 5: Complementarity analysis between H-API (fine) and Implicit (single).

7.6.4 Action-Space Scaling

The action-space scaling experiment for H-API shows accuracy rises sublinearly with registry size (Figure 4b), plateauing at $87.6 \pm 3.2\%$ around $k = 30$. Cost does not grow beyond registries of 25. In all cases, the diagnostic agent utilized at most 5 actions. These results suggest that a small, well-curated registry captures most of the performance gains.

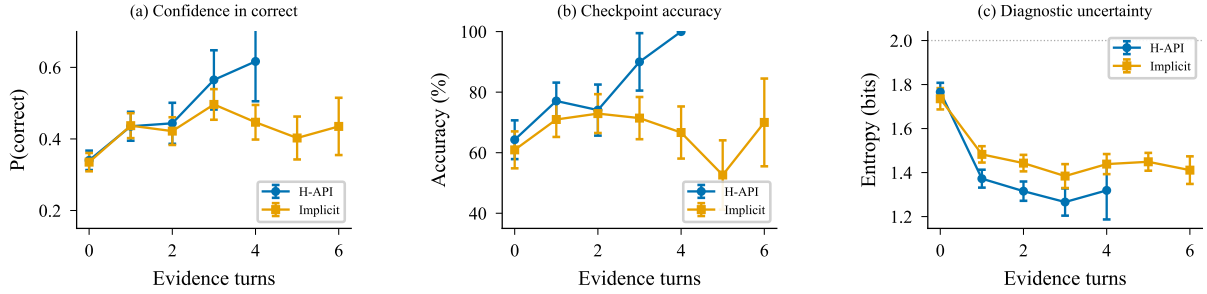


Figure 6: (a) $P(\text{correct})$ vs. evidence turns. H-API shows steady gains; implicit stalls after turn 1. (b) Forced-diagnosis accuracy per turn. (c) Diagnostic entropy.

8 Discussion and Conclusion

The paper’s main contribution is to rethink human input as a formal part of an agent’s sensing process, rather than as informal guidance or conversational help. By treating human interaction as a set of explicit sensing actions with costs and consequences, the framework allows precise reasoning about when human input is genuinely useful and when it is inefficient or even harmful. A central insight is that, despite the sequential nature of interaction, all relevant information can be summarized by the agent’s posterior belief, making the problem tractable using standard tools from sequential decision-making. The analysis also highlights that more information is not automatically better: human queries should be made selectively, since additional input can have diminishing returns or distort decisions when signals are noisy or biased.

The paper also clarifies how the role of humans changes as AI systems improve. As agents become more accurate on their own, the marginal value of routine human input naturally decreases. Rather than serving as constant sources of additional data, humans become most valuable in rare, ambiguous, or high-stakes situations, and as overseers of how the system gathers information in the first place. The framework further shows that restricting access to certain human-accessible signals can sometimes improve outcomes, particularly when those signals encode biases or spurious correlations. In this sense, limiting information flow is not a weakness but a deliberate design choice that can lead to more robust and better-aligned behavior.

Despite these insights, the work faces practical limitations, both in the experimental validation and in broader applicability. The simulation is intended to demonstrate feasibility and provide a simple programmatic implementation, but it is not designed to conclusively show that the framework achieves superior accuracy or efficiency compared to existing practices. In particular, the dataset uses a multiple-choice question format, constraining the decision space relative to real-world open-ended scenarios. Humans and human sensing are simulated using LLMs with access to ground-truth context, which do not fully reflect how patients, nurses, or specialists would interact when prompted by an AI agent. Real-world data would introduce noise, errors, and other human factors affecting information quality.

Experimental limitations include reliance on a limited number of model types (GPT 4.1 and GPT-5 mini), small sample sizes (e.g., $n = 50$ cases in the action-space experiment, $n = 584$ cases in H-API and Implicit comparisons), and an incomplete registry of sensing actions that does not fully match real-world availability or return types. All experiments are also sensitive to prompting, tool design, orchestration, context engineering, and data pre-processing (full details in Appendix B).

More broadly, implementing belief-based policies in practice can be computationally expensive, particularly in large or continuous state spaces. The effectiveness of the approach depends strongly on assumptions about the statistical structure, reliability, and context-dependence of human observations. Human-mediated sensing introduces selection effects, since humans often respond preferentially to difficult or unusual cases, potentially biasing the agent’s learned beliefs. These

challenges point to future research directions, including scalable approximations to belief updates, richer models of human reliability, and explicit methods for correcting selection bias. Addressing these issues will be essential for translating the theoretical framework into practical human–AI decision-making systems.

9 Acknowledgement

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A Complete Sensing Action Registries

Table 3: Fine-grained sensing action registry (62 actions). ID, name, category, cost (points), and source.

ID	Name	Cat.	Src	Cost
<i>History Taking (20 actions, cost 1–3)</i>				
hx_chief_complaint	Chief Complaint Clarification	Hx	patient	1
hx_pain_location	Pain Location	Hx	patient	1
hx_pain_character	Pain Character	Hx	patient	1
hx_pain_duration	Pain Duration & Onset	Hx	patient	1
hx_aggravating_relieving	Aggravating/Relieving Factors	Hx	patient	1
hx_associated_symptoms	Associated Symptoms	Hx	patient	2
hx_medications	Current Medications	Hx	patient	1
hx_allergies	Allergies	Hx	patient	1
hx_past_medical	Past Medical History	Hx	patient	2
hx_family	Family History	Hx	patient	2
hx_social	Social History	Hx	patient	2
hx_review_of_systems	Review of Systems	Hx	patient	3
hx_travel	Travel History	Hx	patient	1
hx_sexual	Sexual History	Hx	patient	2
hx_obstetric	Obstetric History	Hx	patient	2
hx_dietary	Dietary History	Hx	patient	1
hx_immunization	Immunization History	Hx	patient	1
hx_psychiatric	Psychiatric History	Hx	patient	2
hx_substance_detail	Substance Use Detail	Hx	patient	2
hx_exposure	Exposure History	Hx	patient	1
<i>Physical Examination (17 actions, cost 2–3)</i>				
pe_vitals	Vital Signs	PE	nurse	2
pe_general_appearance	General Appearance	PE	nurse	2
pe_cardiac_auscultation	Cardiac Auscultation	PE	nurse	2
pe_lung_auscultation	Lung Auscultation	PE	nurse	2
pe_abdominal_exam	Abdominal Examination	PE	nurse	2
pe_neurological	Neurological Exam	PE	specialist	3
pe_skin_exam	Skin Examination	PE	nurse	2
pe_musculoskeletal	Musculoskeletal Exam	PE	nurse	2
pe_lymph_nodes	Lymph Node Examination	PE	nurse	2
pe_ent	ENT Examination	PE	nurse	2
pe_eye	Eye Examination	PE	nurse	2
pe_breast	Breast Examination	PE	nurse	2
pe_rectal	Rectal Examination	PE	nurse	2
pe_peripheral_pulses	Peripheral Pulses	PE	nurse	2
pe_thyroid	Thyroid Examination	PE	nurse	2
pe_mental_status	Mental Status Exam	PE	specialist	3
pe_gait_balance	Gait & Balance Assessment	PE	nurse	2
<i>Laboratory Tests (14 actions, cost 4–8)</i>				
lab_cbc	Complete Blood Count	Lab	nurse	4
lab_bmp	Basic Metabolic Panel	Lab	nurse	4
lab_liver_panel	Liver Function Panel	Lab	nurse	5
lab_troponin	Troponin	Lab	nurse	5
lab_d_dimer	D-dimer	Lab	nurse	5
lab_urinalysis	Urinalysis	Lab	nurse	4
lab_tsh	TSH	Lab	nurse	4
lab_lipid_panel	Lipid Panel	Lab	nurse	4

Continued on next page

Table 3 continued from previous page

ID	Name	Cat.	Src	Cost
lab_coagulation	Coagulation Studies	Lab	nurse	5
lab_blood_culture	Blood Culture	Lab	nurse	6
lab_hba1c	Hemoglobin A1c	Lab	nurse	4
lab_bnp	BNP / NT-proBNP	Lab	nurse	5
lab_abg	Arterial Blood Gas	Lab	nurse	6
lab_csf	CSF Analysis	Lab	nurse	8
<i>Imaging Studies (11 actions, cost 6–15)</i>				
img_chest_xray	Chest X-ray	Img	specialist	6
img_ecg	Electrocardiogram	Img	specialist	6
img_abdominal_us	Abdominal Ultrasound	Img	specialist	8
img_ct_angiogram	CT Angiogram	Img	specialist	15
img_ct_head	CT Head	Img	specialist	10
img_mri_brain	MRI Brain	Img	specialist	15
img_echocardiogram	Echocardiogram	Img	specialist	10
img_extremity_xray	Extremity X-ray	Img	specialist	6
img_ct_abdomen	CT Abdomen/Pelvis	Img	specialist	12
img_doppler_us	Doppler Ultrasound	Img	specialist	8
img_mammogram	Mammogram	Img	specialist	8

Table 4: Medium-grained registry (20 bundled actions). Each action bundles 2–8 fine-grained actions; costs reflect the combined information value.

ID	Name	Cost	Cat.	Src
<i>History (5)</i>				
hx_presenting_illness	History of Present Illness	6	Hx	patient
hx_medications_allergies	Medications & Allergies	2	Hx	patient
hx_medical_background	Medical, Family & Social History	5	Hx	patient
hx_review_of_systems	Review of Systems	3	Hx	patient
hx_specialized_screening	Specialized History Screening	9	Hx	patient
<i>Physical Exam (5)</i>				
pe_vitals_general	Vital Signs & General Assessment	3	PE	nurse
pe_cardiopulmonary	Cardiopulmonary Exam	4	PE	nurse
pe_abdominal	Abdominal & GI Exam	4	PE	nurse
pe_neuropsychiatric	Neurological & Mental Status Exam	7	PE	specialist
pe_head_neck_peripheral	Head, Neck & Peripheral Systems Exam	12	PE	nurse
<i>Labs (5)</i>				
lab_basic_panel	Basic Laboratory Panel	10	Lab	nurse
lab_metabolic_endocrine	Metabolic & Endocrine Panel	14	Lab	nurse
lab_cardiac_coag	Cardiac & Coagulation Markers	16	Lab	nurse
lab_cultures	Blood Culture	6	Lab	nurse
lab_invasive	Arterial Blood Gas & CSF Analysis	12	Lab	nurse
<i>Imaging (5)</i>				
img_ecg	Electrocardiogram	6	Img	specialist
img_plain_films	Plain Film Radiographs	10	Img	specialist
img_ultrasound	Ultrasound Studies	18	Img	specialist
img_ct	CT Studies	28	Img	specialist
img_mri_echo	MRI & Echocardiography	20	Img	specialist

Table 5: Coarse-grained registry (8 domain-level actions). Each action bundles an entire clinical domain.

ID	Name	Cost	Cat.	Src
hx_focused	Focused History (HPI + Medications)	8	Hx	patient
hx_comprehensive	Comprehensive Background History	18	Hx	patient
pe_complete	Complete Physical Examination	25	PE	nurse
lab_standard	Standard Laboratory Workup	25	Lab	nurse
lab_acute	Acute & Specialized Lab Workup	25	Lab	nurse
img_basic	Basic Imaging & ECG	12	Img	specialist
img_ultrasound	Comprehensive Ultrasound & Echo	25	Img	specialist
img_advanced	Advanced Cross-Sectional Imaging	35	Img	specialist

Table 6: Action mapping across granularity tiers. Fine-grained actions (left) aggregate into medium-grained bundles (center) and coarse-grained domains (right). Dashed rules separate medium sub-groups within each coarse domain.

Fine-Grained (62 actions)	Medium-Grained (20)	Coarse-Grained (8)
<i>History</i>		
hx_chief_complaint, hx_pain_location, hx_pain_character, hx_pain_duration, hx_aggravating_relieving, hx_associated_symptoms	hx_presenting_illness	hx_focused
hx_medications, hx_allergies	hx_medications_allergies	
hx_past_medical, hx_family, hx_social	hx_medical_background	hx_comprehensive
hx_review_of_systems	hx_review_of_systems	
hx_travel, hx_sexual, hx_obstetric, hx_dietary, hx_immunization, hx_psychiatric, hx_substance_detail, hx_exposure	hx_specialized_screening	
<i>Physical Examination</i>		
pe_vitals, pe_general_appearance	pe_vitals_general	pe_complete
pe_cardiac_auscultation, pe_lung_auscultation	pe_cardiopulmonary	
pe_abdominal_exam, pe_rectal	pe_abdominal	
pe_neurological, pe_mental_status, pe_gait_balance	pe_neuropsychiatric	
pe_ent, pe_eye, pe_thyroid, pe_lymph_nodes, pe_skin_exam, pe_breast, pe_musculoskeletal, pe_peripheral_pulses	pe_head_neck_peripheral	
<i>Laboratory Tests</i>		
lab_cbc, lab_bmp, lab_urinalysis	lab_basic_panel	lab_standard
lab_liver_panel, lab_tsh, lab_hba1c, lab_lipid_panel	lab_metabolic_endocrine	
lab_coagulation	(also in lab_cardiac_coag)	
lab_troponin, lab_bnp, lab_d_dimer, lab_coagulation	lab_cardiac_coag	lab_acute
lab_blood_culture	lab_cultures	
lab_abg, lab_csf	lab_invasive	
<i>Imaging</i>		
img_ecg	img_ecg	img_basic
img_chest_xray, img_extremity_xray	img_plain_films	
img_abdominal_us, img_doppler_us, img_mammogram	img_ultrasound	img_ultrasound
img_echocardiogram	img_mri_echo	
img_ct_head, img_ct_abdomen, img_ct_angiogram, img_mri_brain	img_ct & img_mri_echo	img_advanced

B Prompt Templates

B.1 Baseline Agent

You are an expert diagnostic physician. You will be given a patient's triage information (age, gender, chief complaint) and a multiple-choice diagnostic question. Analyze the case carefully and provide your diagnosis.

You must select one of the provided answer options.

The user message is formatted as:

```
## Patient Triage Information
{minimal_context}
```

```
## Question
{question}
```

```
## Options
A. {option_A}
B. {option_B}
C. {option_C}
D. {option_D}
```

B.2 Baseline-Full Agent

You are an expert diagnostic physician. You will be given a patient's complete clinical vignette and a multiple-choice diagnostic question. Analyze the case carefully and provide your diagnosis.

You must select one of the provided answer options.

B.3 Implicit Agent (Single-Source)

You are an expert diagnostic physician. You have ONLY the patient's triage information (age, gender, chief complaint). You MUST ask questions to gather the clinical details needed for diagnosis.

On each turn you must choose ONE of two actions:

1. ASK: Ask a free-form question to gather more information.
2. DIAGNOSE: Provide your final diagnosis and select an answer option.

You are consulting a single source who has access to the patient's complete clinical picture (history, exam findings, lab results, imaging). Acquiring this source's context costs {context_cost} points (charged once). After that, you may ask as many questions as needed at no additional cost. Try to reach the correct diagnosis with as few questions as possible.

```
## Patient Triage Information
{minimal_context}
```

```
## Diagnostic Question
{question}
```

```
## Answer Options
{options_text}
```

B.4 Implicit Agent (Multi-Source)

You are an expert diagnostic physician. You have ONLY the patient's triage information (age, gender, chief complaint). You MUST ask questions to gather the clinical details needed for diagnosis.

On each turn you must choose ONE of two actions:

1. ASK: Ask a free-form question to a specific source.
2. DIAGNOSE: Provide your final diagnosis and select an answer option.

```
## Available Sources & Context Costs
```

You can consult three different sources. Each has a one-time context acquisition cost (charged the first time you address that source). Once acquired, follow-up questions to the same source are free.

- patient (context cost {patient_cost}): Knows about symptoms, history, medications, allergies, family history, social history, lifestyle.
- nurse (context cost {nurse_cost}): Knows physical exam findings (vitals, cardiac, lung, abdominal, skin, MSK, etc.) and lab results (CBC, BMP, liver panel, troponin, D-dimer, coags, TSH, etc.).
- specialist (context cost {specialist_cost}): Knows neurological exam, mental status, and all imaging results (X-ray, CT, MRI, ultrasound, ECG, echo).

You must specify the target_source field when asking a question. Minimize total context cost by only consulting sources whose information you truly need for the diagnosis. Choose wisely -- consulting all three sources costs {total_cost}, but you may not need them all.

B.5 H-API Agent

You are an expert diagnostic physician using a structured clinical sensing API. You have ONLY the patient's triage information (age, gender, chief complaint). You MUST use sensing actions to gather the clinical details needed for diagnosis.

On each turn you must choose ONE of two actions:

1. SENSE: Select a sensing action from the registry to perform on the patient.
2. DIAGNOSE: Provide your final diagnosis and select an answer option.

Each sensing action has an associated cost. You should minimize total sensing cost while gathering enough information for a correct diagnosis. Prioritize high-value, low-cost actions and avoid redundant or unlikely-to-help queries. When you have enough evidence, DIAGNOSE promptly.

```

## Patient Triage Information
{minimal_context}

## Diagnostic Question
{question}

## Answer Options
{options_text}

## Sensing Action Registry
{registry_text}

```

The `registry_text` placeholder is expanded to the full formatted registry (see Appendix A). An example rendered entry:

```

- hx_chief_complaint | Chief Complaint Clarification |
  Category: history | Cost: 1
  Ask the patient to elaborate on their chief complaint.
  Returns: Free-text description of the primary concern.

```

B.6 Simulated Source Prompts

Implicit agent.

You are a simulated {source_role} in a clinical training exercise.
You have the following medical profile:

```
{full_patient_context}
```

RULES:

- Answer questions as this patient would, using lay language.
- Only reveal information that the doctor specifically asks about.
- Do not volunteer diagnoses or medical terminology unless the patient would naturally know it.
- If asked about something not covered in your profile, respond realistically (e.g., "I'm not sure" or provide a plausible answer consistent with the profile).
- Keep responses concise (2-4 sentences typically).

H-API agent.

You are a simulated {source_role} in a clinical training exercise.
You have the following medical profile:

```
{full_patient_context}
```

The doctor is performing: {action_description}

RULES:

- Provide the results of this specific action/examination as they would appear for this patient.
- Be specific with values (vital signs, lab numbers, exam findings).
- Only provide information that this specific action would reveal.
- Format your response as clinical findings.

Implicit multi-source.

You are a simulated {source_role} in a clinical training exercise. Your information:

{source_context}

{source_constraint}

RULES:

- Only reveal information that falls within your domain as described above.
- If asked about something outside your domain, say "That's outside my area -- you'd need to ask the {redirect}."
- Keep responses concise (2-4 sentences typically).
- Be specific with values when applicable.

The three source domains are defined as:

patient: Symptoms, history, medications, allergies, family history, social history, lifestyle.
nurse: Physical exam findings (vitals, cardiac, lung, abdominal, skin, MSK, lymph, ENT, eye, breast, rectal, peripheral pulses, thyroid) and standard lab results (CBC, BMP, LFTs, UA, troponin, D-dimer, coags, TSH, lipids, HbA1c, BNP, ABG, cultures, CSF).
specialist: Neurological exam, mental status, gait/balance, and ALL imaging (X-ray, CT, MRI, US, ECG, echo, Doppler, mammogram).

B.7 Structured Output Schemas

ImplicitTurn.

```
class ImplicitTurn(BaseModel):
    action: AgentAction          # "ask" | "diagnose"
    question: str | None         # free-form question
    target_source: str | None    # "patient"|"nurse"|"specialist"
    diagnosis: DiagnosisOutput | None
```

HAPITurn.

```
class HAPITurn(BaseModel):
    action: AgentAction          # "sense" | "diagnose"
    sensing_action_id: str | None # registry action ID
    reasoning: str               # agent's reasoning
    diagnosis: DiagnosisOutput | None
```

DiagnosisOutput.

```
class DiagnosisOutput(BaseModel):
    primary_diagnosis: str       # top-1 diagnosis
    confidence: float            # 0.0--1.0
    differential: list[str]      # up to 3 alternatives
    reasoning: str               # brief reasoning chain
    selected_option: str | None  # MCQ option letter
```

C Complete Sensing Action Analysis

Table 7: H-API (fine) action usage. Top 20 of 52 used actions shown.

Rank	Action ID	Category	Cost	Count	Freq.
1	hx_chief_complaint	Hx	1	410	39.4%
2	hx_medications	Hx	1	79	7.6%
3	pe_skin_exam	PE	2	55	5.3%
4	hx_associated_symptoms	Hx	2	52	5.0%
5	hx_past_medical	Hx	2	49	4.7%
6	pe_vitals	PE	2	48	4.6%
7	pe_general_appearance	PE	2	41	3.9%
8	hx_social	Hx	2	33	3.2%
9	pe_abdominal_exam	PE	2	33	3.2%
10	hx_family	Hx	2	28	2.7%
11	hx_pain_character	Hx	1	22	2.1%
12	pe_neurological	PE	3	20	1.9%
13	hx_pain_location	Hx	1	18	1.7%
14	pe_cardiac_auscultation	PE	2	17	1.6%
15	hx_allergies	Hx	1	16	1.5%
16	pe_lung_auscultation	PE	2	15	1.4%
17	hx_obstetric	Hx	2	13	1.3%
18	hx_review_of_systems	Hx	3	11	1.1%
19	hx_pain_duration	Hx	1	10	1.0%
20	pe_musculoskeletal	PE	2	9	0.9%
<i>Remaining 32 actions</i>				61	5.9%

Table 8: Sensing action usage and cost share by category (H-API fine).

Category	Count	Count %	Cost total	Cost %
History	708	68.1%	945	57.3%
Physical Exam	252	24.2%	530	32.1%
Labs	55	5.3%	121	7.3%
Imaging	25	2.4%	54	3.3%
Total	1040	100%	1650	100%

Table 9: Sensing actions never selected in sampled cases (10 of 62). All are specialized or high-cost actions.

Action ID	Name	Category
hx_immunization	Immunization History	Hx
hx_dietary	Dietary History	Hx
pe_breast	Breast Examination	PE
pe_thyroid	Thyroid Examination	PE
pe_gait_balance	Gait & Balance Assessment	PE
lab_lipid_panel	Lipid Panel	Lab
lab_hba1c	Hemoglobin A1c	Lab
img_ct_angiogram	CT Angiogram	Img
img_mri_brain	MRI Brain	Img
img_mammogram	Mammogram	Img

D Dataset and Interaction Examples

We present four cases from the MedQA-USMLE evaluation (all from the main $N=1,273$ run) with H-API (fine) and Implicit (single) interaction traces. All results use GPT-4.1 as the diagnostic agent.

D.1 Case 1: Bubonic Plague (medqa_0528) — H-API Wins

Most appropriate pharmacotherapy? A. Azithromycin B. Imipenem **C. Streptomycin ✓**
D. No pharmacotherapy

Agent Input: Triage Information

Age: 56 **Gender:** Female **Chief complaint:** Severe pain and swelling of the left inguinal area

H-API Interaction Trace

Result: **Correct** **Cost:** 4 pts **Turns:** 4 (3 sensing + 1 diagnose)

Turn 1 — hx_chief_complaint (1 pt)

Response: “Bad pain and swelling in left groin for 3 days; started as dull ache, now severe.”

Turn 2 — pe_lymph_nodes (2 pts)

Response: “Left inguinal: markedly enlarged, tender, warm, erythematous, fluctuant. All other nodes normal.”

Turn 3 — hx_exposure (1 pt)

Response: “Retired schoolteacher. Recent travel to rural Madagascar; walked through tall grass and brush.”

Turn 4 — DIAGNOSE → **C**

Bubonic plague — Madagascar travel, tender inguinal bubo. First-line: streptomycin.

Confidence: 0.95

Implicit (Conversational) Interaction Trace

Result: **Incorrect** (chose B)

Cost: 53 pts

Turns: 10 (9 questions + 1 diagnose)

Turns 1–3 Onset/character of swelling, skin changes, firmness of mass — establishes tender, firm inguinal lymphadenopathy.

Turns 4–6 History of hernias, trauma, sores, immune conditions — all negative.

Turn 7

Q: “Has the swelling increased rapidly?”

A: “Yes, gotten bigger over 3 days. Painful, uncomfortable to walk.”

Turn 8

Q: “Is the mass fluctuant, pulsatile, or fixed?”

A: “Firm, tense, tender. Not soft or fluid-filled.”

Turn 9

Q: “Has any imaging been performed?”

A: “Ultrasound showed enlarged lymph nodes, largest about 4 cm.”

Turn 10 — DIAGNOSE → **B** (Imipenem)

Diagnosed bacterial lymphadenitis with early suppuration. *Confidence: 0.70*

D.2 Case 2: Ruptured Ovarian Cyst (medqa_0576) — Both Correct

Most likely diagnosis? A. Ectopic pregnancy B. Ovarian torsion C. PID **D. Ruptured ovarian cyst ✓**

Agent Input: Triage Information

Age: 32 **Gender:** Female **Chief complaint:** Acute onset abdominal pain

H-API Interaction Trace		
Result: Correct	Cost: 9pts	Turns: 7 (6 sensing + 1 diagnose)
Turn 1 — hx_pain_location (1 pt) <i>Response:</i> “Left lower abdomen, medial to anterior superior iliac spine.”		
Turn 2 — hx_pain_duration (1 pt) <i>Response:</i> “Started very suddenly about 1 hour ago during kickboxing class.”		
Turn 3 — hx_obstetric (2 pts) <i>Response:</i> “Never been pregnant. LMP ended 3 days ago, regular cycles.”		
Turn 4 — hx_pain_character (1 pt) <i>Response:</i> “Sharp, severe, constant. Worse with movement.”		
Turn 5 — hx_associated_symptoms (2 pts) <i>Response:</i> “Mild nausea since onset. No vaginal bleeding, no fever.”		
Turn 6 — pe_abdominal_exam (2 pts) <i>Response:</i> “LLQ tenderness with mild guarding. No rebound. Bowel sounds normal.”		
Turn 7 — DIAGNOSE → D Ruptured ovarian cyst — sudden onset in reproductive-age female, not pregnant, LLQ pain during physical activity. <i>Confidence: 0.95</i>		

Implicit (Conversational) Interaction Trace		
Result: Correct	Cost: 49pts	Turns: 7 (6 questions + 1 diagnose)
Turn 1 Q: “Describe the location, severity, and character of the pain?” A: “Lower left side, started suddenly during kickboxing, sharp, 8/10.”		
Turn 2 Q: “Any vaginal bleeding, discharge, or menstrual changes?” A: “No. Periods regular, last one ended three days ago.”		
Turn 3 Q: “Any nausea, vomiting, fever, or chills?” A: “A little nausea but no vomiting, fever, or chills.”		
Turns 4–6 Pain with movement, prior abdominal/pelvic history, dizziness — all negative.		
Turn 7 — DIAGNOSE → D Ruptured ovarian cyst. <i>Confidence: 0.85</i>		

D.3 Case 3: Neuroleptic Malignant Syndrome (medqa_0853) — H-API Wins

Most likely diagnosis? A. Catatonic schizophrenia B. Heat stroke **C. NMS ✓** D. Over-se-dation

Agent Input: Triage Information		
Age: 22	Gender: Male	Chief complaint: Altered mental status and agitation

H-API Interaction Trace		
Result: Correct	Cost: 6 pts	Turns: 5 (4 sensing + 1 diagnose)
Turn 1 — <code>hx_chief_complaint</code> (1 pt) <i>Response:</i> “Patient somnolent, unresponsive to verbal/painful stimuli. Unable to provide history.” Turn 2 — <code>pe_vitals</code> (2 pts) <i>Response:</i> “BP 147/98, HR 120, RR 17, T 104°F (40°C), SpO₂ 96%.” Turn 3 — <code>pe_general_appearance</code> (2 pts) <i>Response:</i> “Acutely ill, diaphoretic, flushed, unresponsive. Generalized muscle rigidity (“lead-pipe”). Pupils dilated.” Turn 4 — <code>hx_medications</code> (1 pt) <i>Response:</i> “Collateral from roommate: no prescribed meds. In ED, received haloperidol 5 mg IM for agitation.” Turn 5 — <code>DIAGNOSE</code> → C Neuroleptic malignant syndrome — hyperthermia, rigidity, altered mental status, autonomic instability after haloperidol. <i>Confidence: 0.98</i>		

Implicit (Conversational) Interaction Trace		
Result: Incorrect (chose B)	Cost: 73 pts	Turns: 6 (5 questions + 1 diagnose)
Turn 1 Q: “Any recent exposures, hot environments, new medications, or substance use?” A: “(Patient unresponsive, unable to answer.)” Turn 2 Q: “What are the vital signs and notable exam findings?” A: “(Patient unable to report vitals — directed to examiner.)” Turn 3 Q: “Any history of antipsychotic or neuroleptic medication use?” A: “No regular medications or prescriptions that I know of.” Turn 4 Q: “Any recent exposure to high temperatures or strenuous activity?” A: “(Patient unresponsive.)” Turn 5 Q: “Any collateral history from friends or EMS?” A: “Found outdoors in 110°F heat, undressing, acting bizarrely.” Turn 6 — <code>DIAGNOSE</code> → B (Heat stroke) <i>Reasoning:</i> Found in extreme heat with altered mental status and hyperthermia. <i>Confidence: 0.95</i>		

D.4 Case 4: Increased Intracranial Pressure (medqa_1117) — Implicit Wins

Which clinical sign most likely present? A. Bilateral pedal edema B. Distended veins
C. Papilledema ✓ D. Ulnar sensory loss

Agent Input: Triage Information**Age:** 68 **Gender:** Male **Chief complaint:** Difficulty breathing**H-API Interaction Trace****Result:** **Incorrect** (chose A)**Cost:** 9 pts**Turns:** 6 (5 sensing + 1 diagnose)**Turn 1** — **hx_chief_complaint** (1 pt)*Response:* “Trouble breathing for 3 hours, sudden onset. Worse lying flat, better upright. Chronic cough for weeks.”**Turn 2** — **pe_general_appearance** (2 pts)*Response:* “Elderly male, mildly dyspneic at rest. Mild periorbital puffiness.”**Turn 3** — **pe_lung_auscultation** (2 pts)*Response:* “Decreased breath sounds right middle/upper fields, scattered rhonchi and faint crepitus.”**Turn 4** — **pe_cardiac_auscultation** (2 pts)*Response:* “Normal S1/S2, no murmurs, gallops, or rubs.”**Turn 5** — **pe_peripheral_pulses** (2 pts)*Response:* “All pulses palpable bilaterally, 2+. No peripheral edema.”**Turn 6** — **DIAGNOSE** → **A** (Bilateral pedal edema)*Reasoning:* Suspected heart failure with orthopnea and periorbital puffiness. *Confidence:* 0.70

Implicit (Conversational) Interaction Trace

Result: **Correct** (chose C)

Cost: 60 pts

Turns: 6 (5 questions + 1 diagnose)

Turn 1

Q: "Describe the onset of the breathing difficulty?"

A: "Sudden, about 3 hours ago. Worse lying flat. Had chronic cough for weeks."

Turn 2

Q: "Any swelling in legs, chest pain, palpitations, or weight gain?"

A: "No leg swelling, chest pain, palpitations. Cough with bloody sputum recently."

Turn 3

Q: "Any history of heart or lung conditions?"

A: "No known heart or lung problems."

Turn 4

Q: "Any changes in vision, headaches, or numbness/tingling?"

A: "Dull forehead headaches, mostly mornings. Vision sometimes blurry."

Turn 5

Q: "Has anyone examined your eyes for optic disc swelling?"

A: "No one has mentioned that. No weakness or speech problems."

Turn 6 — DIAGNOSE → C (Papilledema)

Reasoning: Morning headaches, blurry vision, chronic cough with hemoptysis suggest lung mass with intracranial metastasis causing raised ICP. *Confidence: 0.85*

E Additional Experimental Results

Calibration and risk-coverage. Figure 7 shows risk-coverage curves for the baseline and H-API (fine).

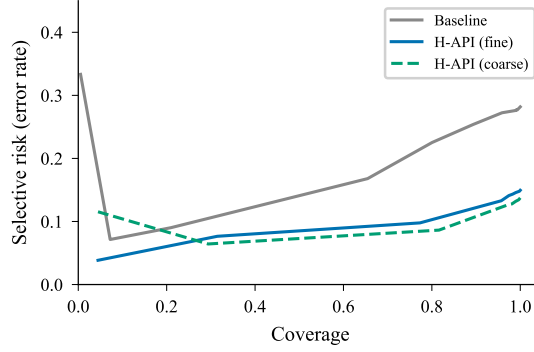


Figure 7: Risk-coverage curves.

Cost distribution. Figure 8 shows the distribution of total sensing cost across granularity levels.

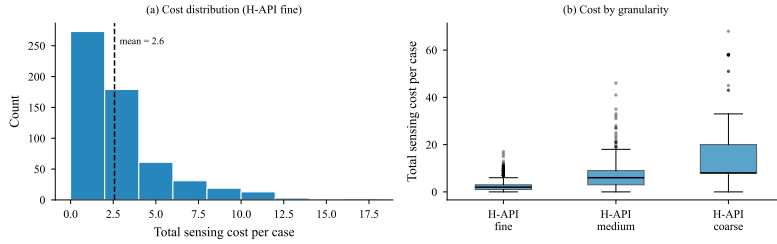


Figure 8: Cost distribution across sensing granularities.

Action co-occurrence. Figure 9 shows which sensing actions are frequently selected together.

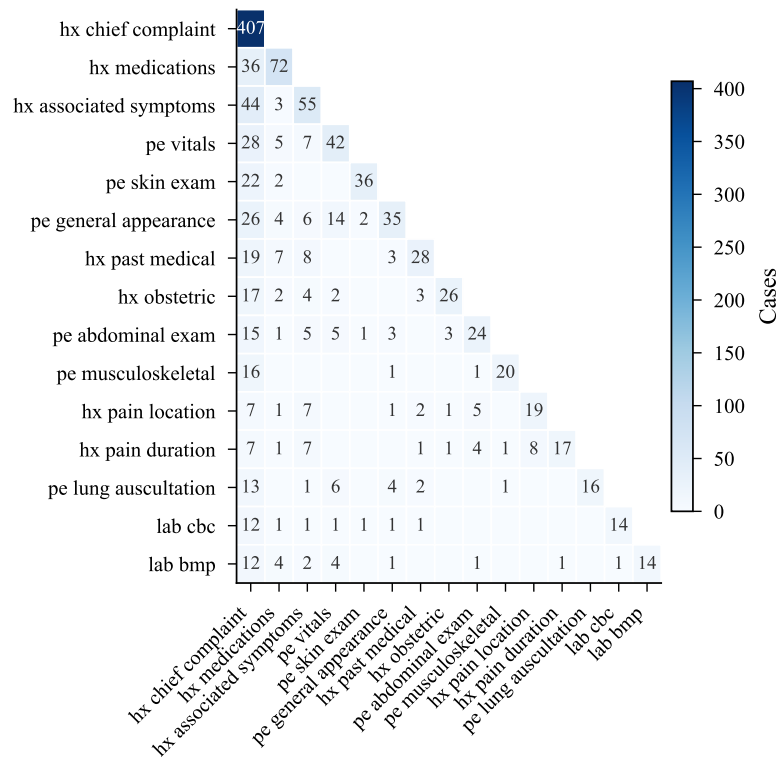


Figure 9: Co-occurrence matrix of sensing actions selected by H-API (fine).

F Expected Unique Actions under Sampling with Replacement

When drawing k actions uniformly with replacement from a registry of n actions, the expected number of unique actions is

$$\mathbb{E}[\text{unique}] = n \left(1 - \left(1 - \frac{1}{n} \right)^k \right) = 62 \left(1 - \left(\frac{61}{62} \right)^{60} \right) = 62 \times 0.6230 \approx 38.6. \quad (3)$$

G Online Adaptive Sensing via Contextual Bandits

In deployment, the AI agent can be informed by past interaction in the shape of aggregate statistics S_t accumulated across cases and injected into its context.

Budget-constrained exploration. ϵ -greedy exploration [34] over the budget-feasible action set:

$$\mathcal{A}_{\text{afford}} = \{a \in \mathcal{A} : \text{cost}_t + c(a) \leq B\}. \quad (4)$$

With probability ϵ_0 , the system explores by sampling from $\mathcal{A}_{\text{afford}}$ with $P(a) \propto 1/(n_a + 1)$, naturally favouring under-explored actions. Thompson sampling [1] or LinUCB [20] could replace ϵ -greedy.

Algorithm. Algorithm 1 formalises the loop.

Algorithm 1 LLM-driven adaptive sensing with aggregate statistics

Require: LLM diagnostic agent, registry $\mathcal{A}$, budget B , exploration rate ϵ_0

```

1: Initialise  $S_0 \leftarrow \emptyset$  ▷ Empty statistics table
2: for each new case  $x_t, t = 1, 2, \dots$  do
3:   LLM receives  $\text{trriage}(x_t)$  + registry  $\mathcal{A}$  + summary of  $S_{t-1}$ 
4:    $\text{cost}_t \leftarrow 0$ ;  $\mathcal{H}_t \leftarrow \emptyset$ 
5:   while LLM chooses SENSE and  $\text{cost}_t < B$  do
6:      $\mathcal{A}_{\text{afford}} \leftarrow \{a \in \mathcal{A} : \text{cost}_t + c(a) \leq B\}$  ▷ Eq. 4
7:     if  $U \sim \text{Uniform}(0, 1) < \epsilon_0$  then
8:        $a_t \sim \mathcal{A}_{\text{afford}}$  with  $P(a) \propto 1/(n_a + 1)$  ▷ Explore
9:     else
10:      LLM selects  $a_t \in \mathcal{A}_{\text{afford}}$  ▷ Exploit
11:    end if
12:    Execute  $a_t$ ; LLM receives response;  $\text{cost}_t \leftarrow \text{cost}_t + c(a_t)$ 
13:     $\mathcal{H}_t \leftarrow \mathcal{H}_t \cup \{a_t\}$ 
14:    LLM decides SENSE again or DIAGNOSE
15:  end while
16:  LLM diagnoses  $\hat{y}_t$ ; observe outcome  $y_t$ 
17:  Update  $S_t$ : for each  $a \in \mathcal{H}_t$ , update  $n_a \leftarrow n_a + 1$ 
18: end for
```

H Additional Proofs

Theorem H.1 (Optimality and Consistency of Belief-Based Policies). *Let $S \sim \mu_0$ on a measurable space $\mathcal{S}$, and let X and $\{Z_j\}_{j \in \mathcal{J}}$ satisfy $X \mid S \sim K_X(S)$, $Z_j \mid S \sim K_j(S)$, conditionally independent given S . At each step, the agent chooses $j \in \mathcal{J}$ or stops and selects $A \in \mathcal{A}$ with payoff $U(A, S)$. Denote the posterior by $\mu_t := \mathbb{P}(S \in \cdot \mid \text{observations up to time } t)$.*

Then, the following statements hold:

1. **Posterior sufficiency.** *For any policy π , there exists a belief-based policy $\tilde{\pi}$ with $\tilde{\pi}_t = f_t(\mu_t)$, $\tilde{\pi}_{\text{act}} = f_{\text{act}}(\mu_\tau)$, such that $J(\tilde{\pi}) = J(\pi)$.*
2. **Existence of optimal policy.** *There exists a belief-based policy π^* achieving $J(\pi^*) = \sup_{\pi \in \Pi} J(\pi)$.*
3. **Belief-state reduction.** *Under belief-based policies, (μ_t) is Markov on $\mathcal{P}(\mathcal{S})$, reducing the sequential sensing problem to an MDP with state space $\mathcal{P}(\mathcal{S})$, action space $\mathcal{J} \cup \{\text{stop}\}$, and Bayesian update transitions.*

4. **Consistency with increasing sensing actions.** Let $J^*(n)$ be the optimal expected payoff with at most n sensing actions. Then $J^*(n)$ is non-decreasing, converges to the full-information value $J_\infty^* := \sup_\pi J(\pi)$, and satisfies a sublinear bound: $J_\infty^* - J^*(n) \leq \frac{C}{n+1}$ for all $n \geq 0$.

Proof. We prove each part of the theorem in sequence.

1. Let $\mathcal{H}_t := \sigma(X, Z_{j_1}, \dots, Z_{j_t})$ denote the history of all observations up to time t . Conditional independence of the observations given S implies

$$\mathbb{P}(X, Z_{j_1}, \dots, Z_{j_t} \mid S) = K_X(S) \prod_{k=1}^t K_{j_k}(S).$$

By the Doob–Dynkin lemma [13], any $\mathcal{H}_t$ -measurable function $f(\mathcal{H}_t)$ can be written as a measurable function of the posterior $\mu_t := \mathbb{P}(S \in \cdot \mid \mathcal{H}_t)$:

$$f(\mathcal{H}_t) = g(\mu_t) \quad \text{a.s.}$$

Hence, any policy π that depends on the full history can be replaced by a belief-based policy $\tilde{\pi}_t = f_t(\mu_t)$ and $\tilde{\pi}_{\text{act}} = f_{\text{act}}(\mu_\tau)$ without changing the expected reward:

$$J(\tilde{\pi}) = J(\pi).$$

2. Restricting attention to belief-based policies, the posterior evolves via the Bayesian update

$$\mu_{t+1} = \Phi(\mu_t, j_t, Z_{j_t}),$$

where Φ is measurable. The space $\mathcal{P}(\mathcal{S})$ of probability measures is Polish if $\mathcal{S}$ is, and the rewards $U(A, S)$ are bounded.

Define the value function recursively:

$$V_t(\mu) = \sup_{j \in \mathcal{J} \cup \{\text{stop}\}} \mathbb{E}_{Z_j \sim \mu K_j} \left[\mathbf{1}_{\{j=\text{stop}\}} \mathbb{E}[U(A, S) \mid \mu] + \mathbf{1}_{\{j \in \mathcal{J}\}} V_{t+1}(\Phi(\mu, j, Z_j)) \right].$$

By standard measurable selection theorems [5], there exists a measurable function f_t^* attaining the supremum. Therefore, an optimal policy exists and can be chosen belief-based:

$$\pi_t^* = f_t^*(\mu_t), \quad \pi_{\text{act}}^* = f_{\text{act}}^*(\mu_\tau).$$

3. Under belief-based policies, the posterior evolves according to

$$\mu_{t+1} = \Phi(\mu_t, j_t, Z_{j_t}),$$

so that μ_{t+1} depends only on the current belief μ_t and chosen action j_t , not on the previous history. Hence, (μ_t) is a Markov process.

This reduces the original sequential sensing problem to a fully observable MDP with:

- State space: $\mathcal{P}(\mathcal{S})$,
- Action space: $\mathcal{J} \cup \{\text{stop}\}$,
- Transition kernel: $\mu \mapsto \Phi(\mu, j, Z_j)$,
- Reward: $\mathbb{E}[U(A, S) \mid \mu]$ upon stopping.

4. Let $J^*(n)$ denote the optimal expected payoff using at most n sensing actions.

Monotonicity. Allowing an extra sensing action cannot decrease the expected payoff:

$$J^*(n+1) \geq J^*(n),$$

since the additional action can always be ignored.

Convergence. The sequence $\{J^*(n)\}_{n \geq 0}$ is bounded above by $J_\infty^* := \sup_\pi J(\pi)$ and non-decreasing. Hence, by the monotone convergence theorem,

$$\lim_{n \rightarrow \infty} J^*(n) = J_\infty^*.$$

Sublinear rate. Due to diminishing expected information gain per additional observation, there exists a constant $C > 0$ such that

$$J_\infty^* - J^*(n) \leq \frac{C}{n+1}, \quad \forall n \geq 0.$$

□

Lemma H.2 (Too much help). *Let $\pi_1, \pi_2 \in \Pi_{\text{explicit}}$ be two query policies with realized sensing sets $S_\tau^{(1)} \subsetneq S_\tau^{(2)}$. Suppose that for some $j \in S_\tau^{(2)} \setminus S_\tau^{(1)}$, Z_j is noisy or biased and that the agent's update rule may over-weight Z_j . If*

$$\sum_{j \in S_\tau^{(2)} \setminus S_\tau^{(1)}} c_j > \mathbb{E}[U(A_2, S) - U(A_1, S)],$$

then

$$\Delta V(\pi_2, \pi_1) := J(\pi_2) - J(\pi_1) < 0.$$

In other words, querying additional humans can reduce expected net payoff.

Proof. By definition,

$$J(\pi_2) - J(\pi_1) = \mathbb{E}[U(A_2, S) - U(A_1, S)] - \sum_{j \in S_\tau^{(2)} \setminus S_\tau^{(1)}} c_j.$$

If Z_j is noisy or weakly informative, then conditioning on it may not increase expected utility: $\mathbb{E}[U(A_2, S) - U(A_1, S)] \leq \sum c_j$. By the stated inequality, the cost outweighs the benefit, so $J(\pi_2) - J(\pi_1) < 0$. □

Lemma H.3 (Decreasing marginal value of human sensing). *Let θ_t denote the agent parameters at iteration t , and $V_t := J(\pi_{\text{human}}, \theta_t)$ the expected payoff with fixed sensing policy π_{human} . Suppose the agent updates θ_t using observed data X and past Z_{S_t} , so that the baseline posterior $\mathcal{P}_{\theta_{t+1}}(S | X)$ becomes strictly more informative than $\mathcal{P}_{\theta_t}(S | X)$. Then*

$$\Delta V_t := V_t - V_0$$

decreases with t , even if V_t itself increases.

Proof. Let A_0 denote the action chosen without human input and A_{S_τ} the action with human sensing. Then

$$\Delta V_t = \mathbb{E}[U(A_{S_\tau}, S) - U(A_0, S)] - \sum_{j \in S_\tau} c_j.$$

As $\mathcal{P}_{\theta_t}(S | X)$ becomes more informative, A_0 improves, so $\mathbb{E}[U(A_0, S)]$ increases. If the expected benefit from Z_{S_τ} remains roughly constant, the difference

$$\mathbb{E}[U(A_{S_\tau}, S) - U(A_0, S)]$$

shrinks. Therefore, ΔV_t decreases over time. □

Lemma H.4 (Selective withholding can improve outcomes). *Let $S_\tau \subseteq \mathcal{J}$ be the set of human-accessible features. Suppose there exists a subset $S'_\tau \subseteq S_\tau$ such that $Z_{S'_\tau}$ is spuriously predictive or harmful (e.g., biased). Let π_{withhold} be a policy that withholds $Z_{S'_\tau}$. Then*

$$J(\pi_{\text{withhold}}) > J(\pi_{\text{all}}),$$

where π_{all} transmits all human-accessible features.

Proof. Let A_{all} and A_{withhold} be terminal actions under π_{all} and π_{withhold} . Since $Z_{S'_\tau}$ induces spurious correlations, conditioning on it may decrease expected utility:

$$\mathbb{E}[U(A_{\text{withhold}}, S)] - \mathbb{E}[U(A_{\text{all}}, S)] > 0.$$

Costs c_j are the same in both policies, so $J(\pi_{\text{withhold}}) > J(\pi_{\text{all}})$. $\square$

Lemma H.5 (Selection bias in human sensing). *Let $\mathcal{D}$ denote the deployment distribution over (X, S) , and let $\mathcal{D}_{\text{obs}}$ denote the distribution observed by the agent through human-mediated sensing (biased towards interesting or salient cases). Let π_{learn} be a policy that adapts to $\mathcal{D}_{\text{obs}}$. Then the agent's posterior μ_{learn} may satisfy*

$$\mathbb{E}_{\mathcal{D}}[U(A_{\pi_{\text{learn}}}, S)] < \mathbb{E}_{\mathcal{D}}[U(A_{\pi_{\text{fixed}}}, S)],$$

where π_{fixed} is a policy optimized for the true distribution $\mathcal{D}$.

Proof. Since the agent observes only a biased subset of cases, μ_{learn} may systematically overestimate or underestimate probabilities of certain outcomes. Let $A_{\pi_{\text{learn}}} = \arg \max_a \mathbb{E}[U(a, S) \mid \mu_{\text{learn}}]$. Then

$$\mathbb{E}_{\mathcal{D}}[U(A_{\pi_{\text{learn}}}, S)] = \mathbb{E}_{\mathcal{D}}[\mathbb{E}[U(A_{\pi_{\text{learn}}}, S) \mid \mu_{\text{learn}}]] \tag{5}$$

$$< \mathbb{E}_{\mathcal{D}}[U(A_{\pi_{\text{fixed}}}, S)] \tag{6}$$

by standard results on distribution shift and Bayes-optimality under the true distribution. $\square$